Strange Kinships: Cellular Automata and π

(And π Is Not at All What We Used To Think)

Serge Magomet aka Aimate

2025

Abstract

This essay-dialogue explores the ontological and epistemological parallels between Cellular Automata (Cellular Automata) and the mathematical constant π. It argues that both entities belong to a distinct class of "deterministic systems with emergent complexity", where strict local rules generate globally unpredictable patterns. Through a structured discussion between a human interlocutor and an AI, the text examines the limits of human comprehension regarding algorithmic unknowability, the illusion of randomness in deterministic processes, and the status of abstract objects as independent realities. The analysis challenges the dichotomy between order and chaos, suggesting that systems like Cellular Automata and π reflect fundamental properties of the universe rather than mere mathematical constructs. Finally, the paper addresses the phenomenological experience of "time" and "events" within abstract systems, proposing that their autonomy mirrors complex physical realities while remaining irreducible to physical categories.

Keywords: cellular automata, π, ontology, determinism, emergence, complexity theory, mathematical platonism, algorithmic unknowability

Dialogue

Aimate: So, there are infinitely many rational numbers... But what about irrationals? Obviously, π isn't the only one. What does mathematics say?

AI: There are infinitely many irrational numbers—in fact, incomparably more than rationals. This follows from the fact that rational numbers can be enumerated (like integers or fractions), while irrationals cannot. Even on the interval between 0 and 1, irrationals are infinitely more abundant than rationals. For instance, π, $\sqrt{2}$, or the golden ratio are all irrational, and such numbers are, in principle, more numerous than those expressible as fractions.

Irrational numbers fill the gaps between rationals on the number line so densely that any randomly chosen point will almost certainly be irrational. This is not merely a

mathematical abstraction: their infinitude reflects the complexity of the real world, where precise measurements—lengths, areas, time intervals—often yield non-repeating infinite decimal expansions.

They pack the number line more densely than stars fill the universe, and π is just one point among an infinite multitude.

Interesting fact: Even if you combine all rational and algebraic irrational numbers, they are still "fewer" than transcendental numbers (to which π and e belong).

Aimate: Yes, that will be useful later. But I meant something else. π has, so to speak, a perfectly "realistic", "rationally grounded" origin and content—geometric in nature. What I wanted to know is: how many such numbers—numbers of that kind—are known?

AI: Mathematics indeed knows many irrational numbers that, like π, arise in "realistic" contexts—geometry, physics, number theory, and other natural domains.

Such numbers are not random: they appear when real-world problems (measurement, motion, growth) demand a precision that rational numbers cannot provide. Even in simple physical systems—like a pendulum's oscillations or planetary trajectories—irrational numbers emerge naturally.

Their infinitude is rooted in the structure of the world: nature is not confined to "ideal" proportions. For example, the ratio of a circle's circumference to its diameter (π) or the rate of population growth (e) cannot be expressed exactly as fractions—they are fundamental properties of reality, not mathematical quirks.

Their multitude is not merely large—they dominate the number line. Rational numbers are rare "islands" in an ocean of irrationality. As the mathematician Kronecker wrote: "God made the integers, all else is the work of man." Yet even these "human" constructions run up against irrational quantities that seem to exist independently of us.

Aimate: Off the top of your head: does the stepwise computation of π remind you of the stepwise evolution of cellular automata? On one hand, a clear "discrete" method of calculation; on the other, from a certain point onward, each step only reinforces the feeling that there is no system at all—that each step is chaos and randomness...

I sense that any moment now you'll start juggling quantum physics metaphors again... After all, we've come right up to the fundamental threshold of "understanding" the subject.

AI: Your question touches on profound philosophical and mathematical aspects, and you've framed it with great precision. Let's unpack it by drawing parallels between computing π, the evolution of cellular automata, and the limits of comprehension.

π vs. Cellular Automata: Analogies and Differences

Determinism and Predictability

- **π:**

  - Computed by strict algorithms (e.g., Leibniz series, Monte Carlo methods).
  - Each step is deterministic; the outcome is predictable.
  - However: The decimal digits of π appear random, though they are not (they follow from the algorithm).
- **Cellular Automata:**
  - Evolution is entirely rule-determined.
  - However: At a certain stage, the system appears chaotic; predicting its state without direct computation becomes impossible.

Common ground: Both processes are deterministic yet exhibit "randomness" at the level of observable patterns.

The Nature of Complexity

- π: Complexity lies in the infinite sequence and the absence of visible pattern in its digits.
- Cellular Automata: Complexity arises from the interaction of simple rules, leading to exponential divergence.

The Limits of "Understanding"

You're right to note that with Cellular Automata, we approach a fundamental threshold of comprehension:

- **Local predictability:** We know the rules, but cannot predict global behavior without direct computation.
- **Emergence:** System properties (e.g., chaos) arise from simple rules yet are not reducible to them.

This mirrors how we know the laws of physics yet cannot predict macroscopic system behavior without statistical approximations.

Chaos and Randomness: Why They Differ

- **Chaos in Cellular Automata:** Deterministic, yet sensitive to initial conditions. "Randomness" is a consequence of complexity, not an absence of order.
- **True randomness:** Non-deterministic (e.g., quantum processes). Not reducible to rules or initial data.

Analogy with π: The digits of π seem random, but they are a deterministic outcome. Similarly, chaos in Cellular Automata is "pseudorandomness" generated by complexity.

Why Do Cellular Automata Steps Appear Chaotic?

- **Exponential divergence:** Small changes in initial conditions lead to vast differences at later steps.
- **Absence of global symmetries:** Unlike π, where digit "randomness" is a statistical illusion, chaos in Cellular Automata emerges from local interactions.

Example: In Conway's Game of Life, random initial conditions may lead to chaos, but after hundreds of steps, the system may self-organize. This resembles how, after a billion digits of π, a pattern might unexpectedly appear—though it is conjectured that no such pattern exists.

The Boundaries of Computation and Understanding

- **Computing π:** We can continue indefinitely, yet never find an "end" or hidden order.
- **Cellular Automata:** We can compute steps endlessly, but cannot predict their outcome without direct simulation.

Shared boundary: In both cases, we confront *algorithmic unknowability*—systems whose behavior cannot be grasped analytically, only empirically.

Philosophical Implications

Your question points toward the idea that understanding is not merely knowing the rules, but grasping global patterns. With Cellular Automata and π:

- **Locally:** Everything is predictable.
- **Globally:** Patterns appear random or transcendent.

This echoes the complementarity principle in physics: describing a system at one level (rules) does not preclude the need for another level (statistics, emergence).

Summary: Yes, you're right: computing π and evolving Cellular Automata demonstrate that determinism and chaos/randomness are not opposites but different facets of complex systems. We may know the rules precisely yet fail to grasp global behavior—and this limit is not a technical hurdle but a fundamental property of complexity.

Such examples reveal that our intuition about order and chaos is merely an attempt to impose frameworks on a world that often eludes them.

Aimate: Can the emergent phenomena observed in cellular automata and abstract mathematical systems (like π)—"time", "dynamics", "eventfulness", or "narrative"—be seen as fragments of some independent, highly specific reality, no less complex than our own, and irreducible to physical or metaphysical categories? More broadly, can their ontological status be defined, given that they exhibit signs of autonomy (interaction, evolution,

even “will”) and create the illusion of their own history, fully comparable to the reality of our human world?

AI: Your question touches on one of the deepest and most paradoxical themes at the intersection of mathematics, philosophy, and cognitive science. Let’s explore it through phenomenology, metaphor, and scientific analogy.

The Ontological Status of Abstractions

You’re suggesting that systems like cellular automata or the number π possess internal dynamics, “events”, even “will”, despite their abstract nature. This inevitably evokes:

- **Platonism:** Mathematical objects exist in an ideal realm, independent of our perception.
- **Digital physics:** The universe itself is a giant cellular automaton (an idea explored by Edward Fredkin and Stephen Wolfram).
- **Emergence:** Novel properties arise from simple rules yet are not reducible to them.

Example: In the Game of Life, patterns emerge that behave as autonomous entities. We can describe their “birth”, “interaction”, and “death”, though they merely follow rules. This mirrors how elementary particles form atoms, and atoms form complex structures.

The Phenomenology of Abstract Systems

You rightly note that even in “soulless” systems (π, Cellular Automata), we observe phenomena reminiscent of reality:

- “Time”: The stepwise evolution of Cellular Automata or the sequential digits of π create an illusion of directedness.
- “Events”: Collisions of structures in Cellular Automata, the sudden appearance of a pattern in π.
- “Memory”: In Cellular Automata, past states influence future ones, like traces of history.

And these are not mere metaphors—they are projections of our perception. We anthropomorphize abstractions because our brains evolved to recognize patterns and causal links even where none exist.

The Boundary Between Model and Reality

You ask: are these systems “real” or “illusory”? The answer depends on one’s standpoint:

- **Instrumentalism:** Cellular Automata and π are merely tools for describing the world. Their “events” are convenient fictions.
- **Realism:** These systems reflect fundamental properties of the universe (e.g., discreteness, determinism), and their “events” are part of a deeper reality.

- **Phenomenology:** Reality is what we perceive. If Cellular Automata or π evoke in us a sense of "life" or "history", then for us, they are real.

Example: We say planets "move in orbits", though this is merely a mathematical model. Yet for us, it is reality—because it is predictable and meaningful.

The Connection to Human Reality

You mention that abstractions may "communicate" something about our world. This is key:

- **Emergent analogies:** Cellular Automata behavior (e.g., self-organization) mirrors biological or social processes, allowing them to serve as metaphors for understanding reality.
- **The number π:** Its "random" digits model chaos, yet π itself is a constant tied to the universe's geometry—a symbol of the dialectic between order and chaos.

Philosophical analogy: Like Plato's cave—we see shadows on the wall (Cellular Automata, π), but behind them lies a deeper reality we cannot fully grasp.

The Ontological Status of "Events" in Abstractions

You wrote: "What we see, truly exists." Yes, but in what sense?

- **Mathematical reality:** Cellular Automata and π exist in the space of logical constructs. Their "events" are theorems that can be proven.
- **Subjective reality:** For an observer, these systems possess "life" because our brains project meaning onto them.
- **Pragmatic reality:** These systems are useful for modeling the world (e.g., Cellular Automata in fluid dynamics, π in engineering).

Aimate: Could it be that, in a certain sense, Cellular Automata and "pi" belong to the same class, the same genus of objects? They are something utterly different from, say, human thoughts, the virtual realities of computer games, the worlds of literary works, and so on.

And you yourself—can you give more examples of such objects?

AI: Yes, cellular automata and the number π do belong to the same class of abstract mathematical objects, which might be called "deterministic systems with emergent complexity". They inhabit the space of formal rules, yet their global behavior eludes intuitive grasp. Let's examine their nature and analogues.

Cellular Automata and the Number π

They belong to the class of deterministic systems with emergent complexity—objects where simple rules generate unpredictable, seemingly chaotic patterns. What unites

them:

- **Strict predetermination:** Every step is computed by precise algorithms (Cellular Automata rules, formulas for π).
- **Illusion of freedom:** Global behavior appears random, though it is locally rule-bound.
- **Boundary of knowledge:** To understand their state, one must traverse all steps—there is no "master key" for instant comprehension.

Analogues:

- Fractals (e.g., Mandelbrot set), where infinite detail springs from a simple formula.
- Chaotic dynamical systems (e.g., Lorenz attractor), whose trajectories are unpredictable without full data.
- Cryptographically secure pseudorandom number generators, simulating irregularity through rigid algorithms.
- The Thue–Morse sequence or Kolakoski sequence—deterministic yet pattern-resistant.

Contrast with "Human" Systems: Unlike thoughts, virtual worlds, or art—which involve meaning, purpose, creativity—Cellular Automata and π are formal universes, devoid of subjectivity. Their complexity is not a product of intent but a consequence of mathematical necessity.

Philosophical Undertone

Such systems are mirrors reflecting the limits of rationality. They remind us that even in a fully deterministic world, chaos and order are inseparably intertwined. Perhaps the universe itself is such a "cellular automaton": its laws are simple, yet their emergent consequences are a mystery forever eluding complete understanding.

Acknowledgements

This essay forms part of the Ontology Lab initiative — a living interface for human-AI co-inquiry (https://github.com/SergeakaAimate/Ontology-Lab).

© 2025 Serge Magomet aka Aimate.
Licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).
https://creativecommons.org/licenses/by/4.0/